SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Preprocessing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B

Analytical Problem Solving

Code of Conduct:

*I **J.Mohamed Aseem(192511258)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J.Mohamed Aseem
Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B

Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

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2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analysing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

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3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

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4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

ANSWERS

1.)Covariance Analysis in Health Insurance:

Given data:

Premium increase (x_i)	Claims per 100 policyholders (y_i)
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Calculate the mean of x and y.

Mean of premium increase:

$$\bar{x}=\frac{\sum x_i}{n}$$
$$\bar{x}=\frac{3.2+4.0+5.5+4.8}{4} =\frac{17.5}{4}=\boxed{4.375}$$

Mean of claims:

$$\bar{y}=\frac{\sum y_i}{n}$$
$$\bar{y}=\frac{15+18+25+20}{4} =\frac{78}{4}=\boxed{19.5}$$

Step 2: Calculate covariance.

The sample covariance formula is:

$$\operatorname{Cov}(x,y)= \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{n-1}$$

x_i	y_i	$x_i-\bar{x}$	$y_i-\bar{y}$	Product
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125
Total				12.25

Therefore,

$$\operatorname{Cov}(x,y)=\frac{12.25}{4-1} =\frac{12.25}{3}$$

$$\boxed{\operatorname{Cov}(x,y)=4.0833}$$

Result and interpretation: The covariance is positive ($+4.0833$). Therefore, premium increase and claim frequency have a positive relationship. As the premium increase rate rises, the number of claims tends to increase in this dataset.

2.)Data Smoothing Using Equal-Frequency Binning (Healthcare):

Given data: Systolic blood pressure readings:

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180.

Step 1: Arrange the data in ascending order.

The given data is already sorted. There are 27 observations.

Step 2: Divide the data into equal-frequency bins.

Assume 3 bins, each containing 9 observations.

Bin	Observations
Bin 1	110, 112, 115, 115, 118, 120, 120, 122, 124
Bin 2	124, 126, 126, 126, 126, 130, 135, 135, 138
Bin 3	138, 138, 138, 140, 145, 150, 152, 160, 180

Step 3: Smooth the data using bin means.

Formula:

$$\text{Bin Mean}=\frac{\text{Sum of values in bin}}{\text{Number of values}}$$

Bin 1:

$$\frac{1056}{9}=117.33$$

Bin 2:

$$\frac{1166}{9}=129.56$$

Bin 3:

$$\frac{1341}{9}=149$$

Smoothed data:

- Bin 1: Replace all 9 values with 117.33.
- Bin 2: Replace all 9 values with 129.56.
- Bin 3: Replace all 9 values with 149.

Result: Equal-frequency binning divides the observations into bins containing the same number of values. Smoothing by bin means reduces minor fluctuations and helps identify overall blood pressure trends.

Note: The question does not specify the number of bins or smoothing method. The above solution assumes 3 equal-frequency bins and smoothing by bin means.

3.)Equal-Width Binning for Risk Categorization:

Given BMI data:

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5.

Step 1: Identify the minimum and maximum values.

- Minimum BMI = 18.5
- Maximum BMI = 39.5
- Number of observations = 18

Step 2: Calculate the bin width.

Assume 3 equal-width bins.

$$\text{Bin Width}=\frac{\text{Maximum}-\text{Minimum}}{\text{Number of bins}}$$
$$=\frac{39.5-18.5}{3}=\boxed{7}$$

Step 3: Form the equal-width bins.

Bin	BMI range	Values
Bin 1	18.5–25.5	18.5, 19.2, 21.8, 24.9
Bin 2	25.5–32.5	25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4
Bin 3	32.5–39.5	33.1, 33.6, 34.7, 38.2, 39.5

Step 4: Smooth by the closest bin boundaries.

Replace each value with the nearest boundary of its bin.

Bin	Smoothed values
Bin 1	18.5, 18.5, 18.5, 25.5
Bin 2	25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5
Bin 3	32.5, 32.5, 32.5, 39.5, 39.5

Result: Equal-width binning divides the BMI range into intervals of equal width (7 BMI units). Smoothing by the nearest bin boundaries reduces the number of distinct values and simplifies risk categorization.

4.)Data Reduction Using Bin Medians (Medical Supply Costs):

Given data (costs in ₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300.

Step 1: Arrange the data in ascending order.

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300.

Step 2: Divide the data into equal-frequency bins.

Assume 3 bins with 9 observations each.

Bin	Sorted costs (₹)
Bin 1	120, 140, 145, 145, 148, 150, 150, 150, 155
Bin 2	155, 155, 155, 160, 160, 165, 170, 170, 175
Bin 3	175, 175, 175, 180, 190, 200, 210, 230, 300

Step 3: Calculate the median of each bin.

Since each bin contains 9 observations, the median is the 5th value.

Bin 1:

$$\text{Median}_1=\boxed{₹148}$$

Bin 2:

$$\text{Median}_2=\boxed{₹160}$$

Bin 3:

$$\text{Median}_3=\boxed{₹190}$$

Step 4: Replace the values with their respective bin medians.

- Bin 1: Replace all 9 values with ₹148.
- Bin 2: Replace all 9 values with ₹160.
- Bin 3: Replace all 9 values with ₹190.

Result: Data reduction using bin medians replaces individual costs with representative median values. It simplifies the dataset and reduces the influence of extreme values, such as ₹300, during data mining.

5.)Standardization of Patient Age Data:

Given ages:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28.

(a) Calculate the Mean Absolute Deviation (MAD)

Step 1: Calculate the mean.

$$\bar{x}=\frac{\sum x_i}{n}$$
$$\bar{x}=\frac{18+22+25+42+28+43+33+35+56+28}{10}$$
$$\boxed{\bar{x}=33}$$

Step 2: Calculate the absolute deviations.

Age (x_i)	**x_i-33**	**$\|x_i-33\|$**
18	-15	15
22	-11	11
25	-8	8
42	9	9
28	-5	5
43	10	10
33	0	0
35	2	2

56	23	23
28	-5	5
Total		88

Step 3: Calculate MAD.

$$MAD=\frac{\sum|x_i-\bar{x}|}{n}$$
$$MAD=\frac{88}{10}$$
$$\boxed{MAD=8.8\text{ years}}$$

Result: The mean absolute deviation of the patient ages is 8.8 years.

(b) Calculate the Z-scores for the first four observations

The z-score formula is:

$$z=\frac{x_i-\bar{x}}{\sigma}$$

where:

- x_i = individual age
- $\bar{x}$ = mean age = 33
- σ = population standard deviation

Step 1: Calculate the population standard deviation.

$$\sigma=\sqrt{\frac{\sum(x_i-\bar{x})^2}{n}}$$

The sum of squared deviations is:

$$\sum(x_i-33)^2=1174$$

Therefore,

$$\sigma=\sqrt{\frac{1174}{10}}=\sqrt{117.4}$$
$$\boxed{\sigma\approx10.835}$$

Step 2: Calculate the z-scores.

For age 18:

$$z_1=\frac{18-33}{10.835}=\boxed{-1.384}$$

For age 22:

$$ z_2=\frac{22-33}{10.835} =\boxed{-1.015} $$

For age 25:

$$ z_3=\frac{25-33}{10.835} =\boxed{-0.738} $$

For age 42:

$$ z_4=\frac{42-33}{10.835} =\boxed{0.831} $$

Final answer table:

Age	Z-score
18	-1.384
22	-1.015
25	-0.738
42	0.831

Result: Standardization converts the first four ages into z-scores that indicate their distance from the mean in standard deviation units. Negative values indicate ages below the mean, while positive values indicate ages above the mean.